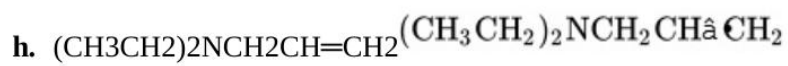
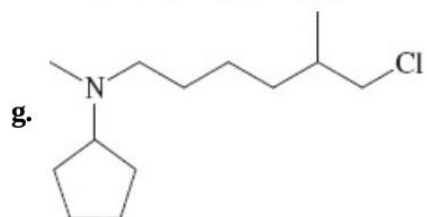
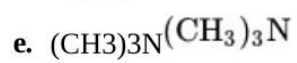
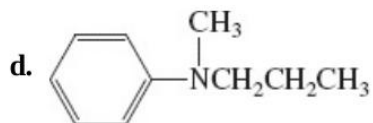
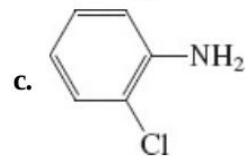
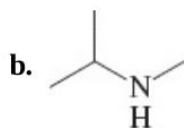
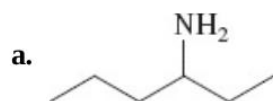


27. Give at least two names for each of the following amines.



28. Give structures that correspond to each of the following names.

a. *N,N*-Dimethyl-3-cyclohexenamine;

b. *N*-ethyl-2-phenylethylamine;

c. 2-aminoethanol;

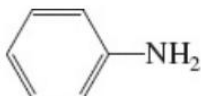
d. *m*-chloroaniline.

30. As mentioned in [Section 21-2](#), the inversion of nitrogen requires a change of hybridization. **(a)** What is the approximate energy difference between pyramidal nitrogen (sp^3 hybridized) and trigonal planar nitrogen (sp^2 hybridized) in ammonia and simple amines? (**Hint:** Refer to the E_a of inversion.) **(b)** Compare the nitrogen atom in ammonia with the carbon atom in each of the following species: methyl cation, methyl radical, and methyl anion. Contrast the most stable geometries and the hybridizations of each of these species. Using fundamental notions of orbital energies and bond strengths, explain the similarities and differences among them.

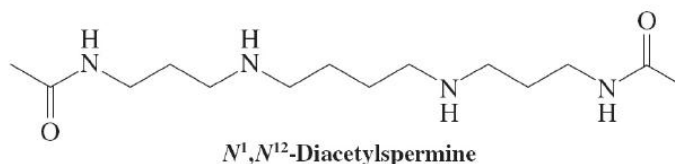
35. In which direction would you expect each of the following equilibria to lie?



36. (a) How would you expect the following classes of compounds to compare with simple primary amines as bases and acids?

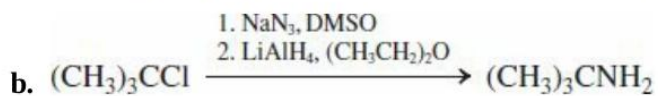
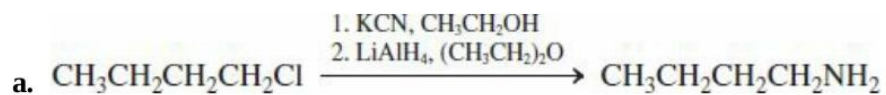
- i. Carboxylic amides; for example, CH_3CONH_2 CH_3CONH_2
- ii. Imides; for example, $\text{CH}_3\text{CONHCOCH}_3$ $\text{CH}_3\text{CONHCOCH}_3$
- iii. Enamines; for example, $\text{CH}_2=\text{CHN}(\text{CH}_3)_2$ $\text{CH}_2=\text{CHN}(\text{CH}_3)_2$
- iv. Anilines; for example, 

(b) The tetraamine derivative shown below has been found to be strongly associated with several human cancers. It is relatively scarce, making up less than 0.5% of naturally occurring polyamines in human serum. However, its concentration rises by some 50% in the urine of patients suffering from non-small cell lung cancer, which is the leading cause of cancer deaths worldwide. The detection of this substance holds promise in permitting early-stage diagnosis, when treatment is likely to be more effective. Recent studies showing elevated levels of this substance in colon cancer patients have revealed that it is not only a cancer biomarker, but also a cancer promoter. Thus, bacterial biofilms in the colon favor accumulation of this compound, and antibiotic treatment to clear the biofilms reduces its concentration to normal, thus paving the way for new approaches to control cancer growth.



At normal physiological pH this substance exists in a doubly protonated form with pK_a values of 10.2 and 10.6. Write the structures that should be present at $\text{pH} = 7$, 10.4, and 12.

39. In the following syntheses of amines, indicate in each case whether the approach will work well, poorly, or not at all. If a method will not work well, explain why.



40. For each synthesis in [Problem 39](#) that does not work well, propose an alternative preparation of the final amine, starting either with the same material or with a material of similar structure and functionality.

41. Give the structures of all possible nitrogen-containing organic products

47. Formulate a detailed mechanism for the Mannich reaction between 2-methylpropanal, formaldehyde, and methanamine shown on [p. 1042](#).

50. Show how each of the following molecules might be synthesized by Mannich or Mannich-like reactions. (**Hint:** Work backward, identifying the bond made in the Mannich reaction.)

